Seroprevalence of SARS-CoV-2 among vaccinated people in Asubiaro General Hospital (Osogbo) and Federal Medical Centre, Owo

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Summary

SARS-CoV-2 vaccines trigger antibodies against the spike protein. Antibodies against the nucleocapsid (N) protein are not triggered by the vaccines used in Nigeria, so they point to natural infection. By measuring both, a blood sample can show whether someone has a vaccine response only, or a vaccine response plus signs of past infection.

This project measured both antibody types by ELISA in 68 archived plasma samples from vaccinated adults at two hospitals, one in Osun State (Asubiaro General Hospital, Osogbo) and one in Ondo State (Federal Medical Centre, Owo). Results were compared by location, sex and age group.

Abstract

SARS-CoV-2, the virus responsible for COVID-19, elicits various immune responses. This study focused on spike receptor-binding domain (SRBD) IgG antibodies, which are elicited by vaccination, and nucleocapsid (N) IgG antibodies, which are elicited by contact with the virus. Antibody patterns were assessed in vaccinated people at Federal Medical Centre, Owo, Ondo State, and Asubiaro General Hospital, Osun State, across age groups, sexes and locations. The findings point to a need for targeted public health interventions and continued vaccination efforts.

Keywords: SARS-CoV-2, immune response, antibodies, SRBD IgG, N IgG, seroprevalence

Methods

- **Participants:** 68 vaccinated adults (aged 18 and over): 33 from Asubiaro General

Hospital and 35 from Federal Medical Centre, Owo. 48 were female and 20 were male. Participants had a negative rapid diagnostic test at enrolment, a sighted vaccination card, and gave written informed consent. Pregnant individuals were excluded.

- **Samples:** archived plasma samples from the GIISER-ARISE project, collected as 5 ml venous blood, separated by centrifugation, transported in cold chain and stored at -20 C.
- **Assay:** direct-format ELISA detecting human IgG against the SARS-CoV-2 spike receptor-binding domain (SRBD) and nucleocapsid (N) proteins, using the ReSARS CoV-2 (SRBD) and (N) ELISA kit. Optical density was read at 450 nm with a 620 to 650 nm reference.
- **Age groups:** young (18 to 29), middle-aged (30 to 50) and older (over 50).
- **Analysis:** Microsoft Excel and GraphPad Prism 9, using unpaired Mann-Whitney tests and ANOVA on optical density values.
- **Ethics:** approved by the Medical Directors of both hospitals, with written informed consent. The work was part of the Africa Rapid Immunosurveillance System for Epidemic Response (ARISE) project.

Results

Antibody patterns (counts from the project)

	Owo male	Asubiaro male	Owo female	Asubiaro female	Total
Anti-spike (SRBD) only	2	1	12	4	19
Anti-nucleocapsid (N) only	1	1	1	2	5
Both	6	9	13	16	44
Total	9	11	26	22	68

Every participant had at least one of the two antibody types: 28% had anti-spike antibodies only, 7% anti-nucleocapsid only, and 65% both.

Antibody levels (optical density)

- Anti-spike IgG levels were similar in the two cohorts (geometric mean OD 1.094 in Owo

and 1.100 in Asubiaro; p = 0.8932).

- Anti-N IgG levels were higher in the Asubiaro cohort (geometric mean OD 1.120 vs 0.8683 in Owo).
- Among participants with both antibodies, geometric mean OD values were higher in Asubiaro (spike 1.575 vs 1.187; N 0.8367 vs 0.6610).

Supplementary reanalysis of the counts

The original percentages were calculated out of all 68 participants. Because 70% of participants were female, comparisons by sex mostly reflect group size. `seroprevalence_reanalysis.py` recalculates every percentage within its own group, with 95% Wilson confidence intervals and Fisher exact tests, using only the counts above.

Finding	Result
Any N antibodies (signal of past infection)	49 of 68, 72% (95% CI 60 to 81)
Any N antibodies, Asubiaro vs Owo	85% vs 60%, OR 3.7 (95% CI 1.0 to 15.1), p = 0.031
Both antibodies, Asubiaro vs Owo	76% vs 54%, OR 2.6 (95% CI 0.8 to 8.6), p = 0.079
Any N antibodies, males vs females	85% vs 67%, p = 0.150
Any N antibodies by age group	Young 50% (4 of 8), middle-aged 67% (31 of 46), older 100% (14 of 14)
Trend across age groups	z = 2.73, p = 0.006

These tests are exploratory and not corrected for multiple comparisons. With groups this small, the confidence intervals are wide.

Discussion

Within these 68 vaccinated people, signs of past infection were common, and they were more frequent in the Asubiaro sample and in older participants. The two sites also differed in the share of people with only a vaccine-type response (40% in Owo vs 15% in Asubiaro). Differences by sex disappear once percentages are calculated within each sex, so the sex differences suggested by the raw counts should be read with care.

Limitations

- **Small, unequal groups.** 68 participants in total, with 8 in the youngest age group and 20 men.
- **Not a population estimate.** Samples came from hospital patients and surrounding communities, not a random sample of the population.
- **Archived samples.** Participants were not followed up, so the study is cross-sectional and cannot say when infection happened or how long antibodies last.
- **No unvaccinated comparison group,** so the study cannot estimate vaccine effectiveness.
- **Assay.** The ELISA kit is for research use, and a study-specific cut-off should be set using seronegative samples.
- **Variants** were not considered.

Files

File	What it is
`antibody_pattern_counts.csv`	Participant counts by location, sex or age group, and antibody pattern (from Tables 3 and 4)
`seroprevalence_reanalysis.py`	Within-group percentages, confidence intervals and tests
`reanalysis_results.csv`	Output of the reanalysis
`antibody_patterns.png`	Antibody patterns by location, sex and age group

Participant-level optical density data are not included in this repository.

Run the reanalysis

```
pip install numpy pandas scipy matplotlib
python seroprevalence_reanalysis.py
```

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How to cite

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